

```
!pip install transformers datasets torch scikit-learn pandas
```

```
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
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Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
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Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.10.0) (4.0.0a1)
Requirement already satisfied: hf-xet<2.0.0,>=1.3.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0.0,>=1.3.0) (1.3.2)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0.0,>=1.3.0) (0.23.0)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0.0,>=1.3.0->transformers) (0.12.5)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.10.1)
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Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch) (3.0.3)
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Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0.0,>=1.3.0) (0.16.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0.0,>=1.3.0) (8.2.1)
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Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0.0,>=1.3.0) (13.9.0)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0.0,>=1.3.0) (0.0.2)
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Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer) (2.19.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0) (0.1.2)
```

```
import torch
from datasets import load_dataset
from transformers import AutoTokenizer, AutoModelForSequenceClassification
from transformers import TrainingArguments, Trainer
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
```

```
dataset = load_dataset("imdb")

train_dataset = dataset["train"]
test_dataset = dataset["test"]
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
README.md:      7.81k/? [00:00<00:00, 310kB/s]

plain_text/train-00000-of-00001.parquet: 100%                21.0M/21.0M [00:00<00:00, 105MB/s]

plain_text/test-00000-of-00001.parquet: 100%                20.5M/20.5M [00:00<00:00, 102MB/s]

plain_text/unsupervised-00000-of-00001.p(...): 100%         42.0M/42.0M [00:00<00:00, 108MB/s]

Generating train split: 100%                                25000/25000 [00:00<00:00, 104804.82 examples/s]

Generating test split: 100%                                25000/25000 [00:00<00:00, 123811.98 examples/s]

Generating unsupervised split: 100%                        50000/50000 [00:00<00:00, 151107.54 examples/s]
```

```

tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")

def tokenize_function(example):
    return tokenizer(example["text"], padding="max_length", truncation=True)

train_dataset = train_dataset.map(tokenize_function, batched=True)
test_dataset = test_dataset.map(tokenize_function, batched=True)
```

```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
config.json: 100%                483/483 [00:00<00:00, 56.8kB/s]

tokenizer_config.json: 100%                48.0/48.0 [00:00<00:00, 5.23kB/s]

vocab.txt:      232k/? [00:00<00:00, 6.10MB/s]

tokenizer.json:   466k/? [00:00<00:00, 15.9MB/s]

Map: 100%                25000/25000 [00:25<00:00, 953.08 examples/s]

Map: 100%                25000/25000 [00:24<00:00, 1090.14 examples/s]
```

```

train_dataset = train_dataset.remove_columns(["text"])
test_dataset = test_dataset.remove_columns(["text"])

train_dataset.set_format("torch")
test_dataset.set_format("torch")
```

```

train_dataset
test_dataset

Dataset({
  features: ['label', 'input_ids', 'token_type_ids', 'attention_mask'],
  num_rows: 25000
})
```

```

model = AutoModelForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=2
)
```

```

model.safetensors: 100%                268M/268M [00:02<00:00, 197MB/s]

Loading weights: 100%                100/100 [00:00<00:00, 459.06it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]

DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key | Status |
-----+-----+
vocab_layer_norm.bias | UNEXPECTED |
vocab_projector.bias | UNEXPECTED |
vocab_transform.weight | UNEXPECTED |
vocab_layer_norm.weight | UNEXPECTED |
vocab_transform.bias | UNEXPECTED |
pre_classifier.weight | MISSING |
classifier.weight | MISSING |
classifier.bias | MISSING |
pre_classifier.bias | MISSING |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream t
```

```
def compute_metrics(eval_pred):
    logits, labels = eval_pred
    predictions = torch.argmax(torch.tensor(logits), axis=1)

    precision, recall, f1, _ = precision_recall_fscore_support(
        labels, predictions, average='binary'
    )

    acc = accuracy_score(labels, predictions)

    return {
        "accuracy": acc,
        "f1": f1,
        "precision": precision,
        "recall": recall
    }
```

```
training_args = TrainingArguments(
    output_dir="./results",
    eval_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=2,
    weight_decay=0.01
)
```

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset.shuffle(seed=42).select(range(5000)),
    eval_dataset=test_dataset.shuffle(seed=42).select(range(1000)),
    compute_metrics=compute_metrics
)
```

```
trainer.evaluate()
trainer.train()
```

 [125/125 04:59]

 [1250/1250 09:53, Epoch 2/2]

Epoch	Training Loss	Validation Loss	Model Preparation Time	Accuracy	F1	Precision	Recall
1	0.360475	0.330513	0.001400	0.892000	0.894531	0.854478	0.938525
2	0.241442	0.358540	0.001400	0.898000	0.898000	0.876953	0.920082

Writing model shards: 100%

1/1 [00:06<00:00, 6.96s/it]

Writing model shards: 100%

1/1 [00:06<00:00, 6.86s/it]

Writing model shards: 100%

1/1 [00:01<00:00, 1.63s/it]

TrainOutput(global_step=1250, training_loss=0.27558659973144534, metrics={'train_runtime': 593.8581, 'train_samples_per_second': 16.839, 'train_steps_per_second': 2.105, 'total_flos': 132467398656000.0, 'train_loss': 0.27558659973144534, 'epoch': 2.0})

```
text = "The movie was fantastic and full of suspense"
```

```
inputs = tokenizer(text, return_tensors="pt").to("cuda")
```

```
outputs = model(**inputs)
```

```
prediction = torch.argmax(outputs.logits)
```

```
print("Positive" if prediction==1 else "Negative")
```

```
Positive
```